CYLINDER HEAD (Continued)

COOLING SYSTEM TESTER METHOD

WARNING: WITH COOLING SYSTEM TESTER IN PLACE, PRESSURE WILL BUILD UP FAST. EXCESSIVE PRESSURE BUILT UP, BY CONTINUOUS ENGINE OPERATION, MUST BE RELEASED TO A SAFE PRESSURE POINT. NEVER PERMIT PRESSURE TO EXCEED 138 kPa (20 psi).

Install Cooling System Tester 7700 or equivalent to pressure cap neck. Start the engine and observe the tester's pressure gauge. If gauge pulsates with every power stroke of a cylinder a combustion pressure leak is evident.

CHEMICAL TEST METHOD

Combustion leaks into the cooling system can also be checked by using Bloc-Chek Kit C-3685-A or equivalent. Perform test following the procedures supplied with the tool kit.

REMOVAL - CYLINDER HEAD

(1) Perform fuel system pressure release procedure **before attempting any repairs.** (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY - SPECIFICATIONS)
(2) Disconnect battery negative cable.
(3) Drain cooling system. (Refer to 7 - COOLING - STANDARD PROCEDURE)
(4) Remove air filter housing and inlet tube.
(5) Remove intake manifold.
(6) Remove heater tube support bracket from cylinder head.
(7) Disconnect radiator upper and heater supply hoses from water outlet connections.
(8) Remove accessory drive belts. (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL)
(9) Raise vehicle and remove exhaust pipe from manifold.
(10) Remove power steering pump and set aside. Do not disconnect lines.
(11) Remove accessory drive bracket
(12) Remove ignition coil and wires from engine.
(13) Disconnect cam sensor and fuel injector wiring connectors.
(14) Remove timing belt and camshaft sprockets. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)
(15) Remove timing belt idler pulley and rear timing belt cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)
(16) Remove cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL)
(17) Remove camshafts (Refer to 9 - ENGINE/CYLINDER HEAD/CAMSHAFT(S) - REMOVAL).

NOTE: Identify rocker arm position to ensure correct re-installation in original position, if reused.

(18) Remove rocker arms. (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARMS - REMOVAL).
(19) Remove cylinder head bolts in REVERSE sequence of tightening.
(20) Remove cylinder head from engine block.
(21) Inspect and clean cylinder head. (Refer to 9 - ENGINE/CYLINDER HEAD - INSPECTION) (Refer to 9 - ENGINE/CYLINDER HEAD - CLEANING)

CLEANING

To ensure engine gasket sealing, proper surface preparation must be performed, especially with the use of aluminum engine components and multi-layer steel cylinder head gaskets.

NOTE: Multi-Layer Steel (MLS) head gaskets require a scratch free sealing surface.

Remove all gasket material from cylinder head and block (Refer to 9 - ENGINE - STANDARD PROCEDURE). Be careful not to gouge or scratch the aluminum head sealing surface.
Clean all engine oil passages.

INSPECTION

(1) Cylinder head must be flat within 0.1 mm (0.004 in.) (Fig. 8).
(2) Inspect camshaft bearing journals for scoring.

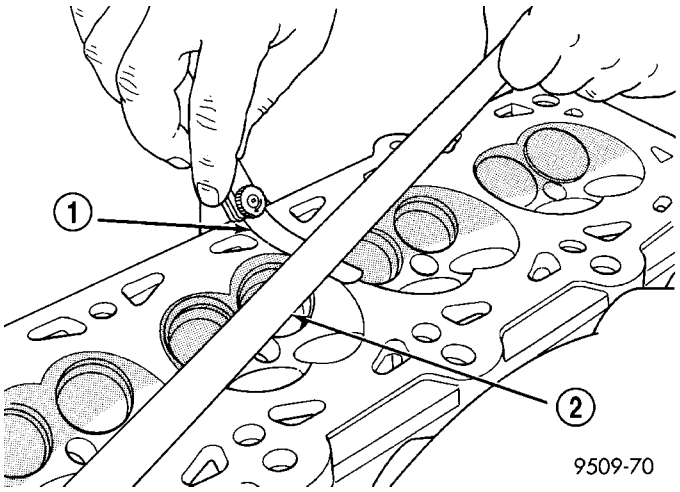

Fig. 8 Checking Cylinder Head Flatness

1 - FEELER GAUGE
2 - STRAIGHT EDGE

(3) Remove carbon and varnish deposits from inside of valve guides with a reliable guide cleaner.